

Rising Waters: A Machine Learning Approach to Flood Prediction

No. of Functional Features Included in the Solution

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Weather Dataset Collection	Collects historical weather and flood-related data required for prediction	Dataset Module	Done	High
2	Data Preprocessing	Handles missing values, outliers, categorical values, and feature scaling	Data Preprocessing Module	Done	High
3	Exploratory Data Analysis	Analyzes weather patterns using visualization and statistical methods	Data Analysis Module	Done	Medium
4	Machine Learning Model Training	Trains classification models such as Decision Tree, Random Forest, KNN, and XGBoost	ML Model Module	Done	High
5	Model Evaluation and Selection	Compares model performance using accuracy, confusion matrix, and classification report	Evaluation Module	Done	High
6	Flood Risk Prediction	Predicts flood chance based on user-provided weather conditions	Prediction Module	Done	High
7	Flask Web Application	Provides user interface for entering data and viewing results	Flask Application Module	Done	High
8	Cloud Deployment Support	Enables future deployment on IBM Cloud for scalability	Deployment Module	In Progress	Medium

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	7
Core / Must-Have Features	Data preprocessing, ML model training, prediction, Flask application
Additional / Nice-to-Have Features	Cloud deployment, real-time weather API integration
Features Tested & Verified	7

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Flask web pages, prediction input form, result display
2	Backend / Logic	4	Data preprocessing, model training, prediction logic, model evaluation
3	Database / Storage	2	Weather dataset storage, trained model storage (floods.save, scaler.save)
4	API / Integration	1	Flask server integration, future weather API support
5	Security / Authentication	0	Not implemented in the current version